

CHP #5

⇒ Arithmetic mean:- (Average)

i) A.M for ungrouped data:-

• If the data set represent a finite population of size N . Then A.M is denoted by " μ " and calculated as:

$$\mu = \frac{\sum X}{N}$$

• If the data set represent a finite sample of size " n ", then denoted and calculated as:-

$$\bar{X} = \frac{\sum x}{n}$$

Understanding by example

1(a) For population size:-

The number of employees at 6 different stores = 6, 8, 7, 8, 9, 10

$N = 6$

$$\mu = \frac{\sum X}{N} = \frac{6+8+7+8+9+10}{6} = \frac{48}{6} = 8$$

Example 2(a)

Date: _____

the daily pocket money received by five first year students = 80, 120, 110, 90, 100

$$n = 5$$

$$\bar{X} = \frac{\sum X}{n} = \frac{80 + 120 + 110 + 90 + 100}{5} = \frac{500}{5} = 100$$

A.M For grouped data

If the values $x_1, x_2, x_3, \dots, x_n$ occur

with frequencies $f_1, f_2, f_3, \dots, f_n$ then A.M is

$$\bar{X} = \frac{\sum f_x}{\sum f} \quad \text{or} \quad \bar{X} = \frac{\sum f_x}{n}$$

where $n = \sum f$ the total frequencies

Example 3(a)

calculate average age of first year student from the following frequencies

Age (year)	No. of students
13	2
14	5
15	13
16	7
17	3

Here age is a variable X and no. of students is frequencies f

X	f	fX
13	2	26
14	5	70

15	13	195
16	7	112
17	3	51
Total	$\Sigma f = 30$	$\Sigma fx = 454$
$\bar{x} = \frac{\Sigma fx}{\Sigma f} = \frac{454}{30} = 15.134 \text{ years}$		

Example 3(b)

Answers:-

$\therefore \bar{x} = \frac{\Sigma fx}{\Sigma f}$ where x is the class mark
 Σf or mid point and f is no.
 of student

So:- Class interval	x	f	fx
100-104	102	5	510
105-109	107	15	1605
110-114	112	20	2240
115-119	117	25	2925
120-124	122	30	3660
125-129	127	20	2540
130-134	132	16	2112
135-139	137	14	1918
140-144	142	5	710
Total		$\Sigma f = 150$	$\Sigma fx = 18220$

$$\bar{X} = \frac{\sum fx}{\sum f} = \frac{18220}{150} = 121.466$$

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The weighted arithmetic mean:-
The mean obtained by assigning each observation a weight that reflect its importance. Such

$$\bar{X}_w = \frac{\sum wx}{\sum w}$$

Example 4(a)

compute weight average

sol:- W (No. of emp) X (Monthly pay) (XW)

4	800	3200
22	45	990
20	100	2000
30	30	900
80	35	2800
300	15	4500
$\sum W = 456$		$\sum 14390$

$$\bar{X}_w = \frac{\sum XW}{\sum W} = \frac{14390}{456} = 31.55$$

Example 4(b)

Subjects	Marks (X)	weights (W)	WX
English	73	4	292
Math	82	3	246

Math	80	3	240
History	67	2	134
Science	62	2	124

Total:- $\Sigma W = 14$ $\Sigma WX = 1036$

i) compute average marks according to the data.

$$\bar{X} = \frac{\Sigma WX}{\Sigma W} = \frac{1036}{14} = 74 \text{ marks}$$

ii) If equal weights are used than arithmetic mean

$$\bar{X} = \frac{\Sigma X}{n} = \frac{73+82+80+67+62}{5} = 72.8$$

Example 6:

X	$f(W)$	WX
38.9	21.5	836.35
41.9	18.8	787.72
42.9	23.8	1021.02
Total	64.1	$\Sigma WX = 2645.09$

Mean Price paid per litre:-

$$\bar{X} = \frac{\Sigma XW}{\Sigma W} = \frac{2645.09}{64.1} = 41.26 \text{ cents/litre}$$

Combined A.M:

If n_1 values have mean $\bar{X}_1$, n_2 values have mean $\bar{X}_2$, ..., n_k

values have mean $\bar{x}_k$ then combined

$$\bar{x}_c = \frac{n_1 x_1 + n_2 x_2 + \dots + n_k x_k}{n_1 + n_2 + n_3 + \dots + n_k} = \frac{\sum_{i=1}^k n_i x_i}{\sum_{i=1}^k n_i}$$

or $\bar{x}_c = \frac{\sum n_i x_i}{\sum n_i}$

Example 7(a)

Sections of mathematics - 3
class have no of students - 30, 35, 40

Averaged = 80, 85, 75

combined mean $\bar{x}_c = ?$

$$\bar{x}_c = \frac{\sum n_i x_i}{\sum n_i} = \frac{n_1 x_1 + n_2 x_2 + n_3 x_3}{n_1 + n_2 + n_3}$$

$$= \frac{30(80) + 35(85) + 40(75)}{30 + 35 + 40}$$

$$= \frac{2400 + 2975 + 3000}{105} = \frac{8375}{105} = 79.76$$

Median:- (ungrouped) data

For odd terms

7, 18, 19, 25, 78

Median = 19

by formula - $\frac{(n+1)}{2}$ th term

total terms = 5

$n=5$

Date: _____

$$\text{Median} = \frac{(5+1)\text{th} + 6\text{th}}{2} = \frac{3\text{th} + 6\text{th}}{2}$$

$$|\text{Median} = 19| \Rightarrow \boxed{\tilde{x} = 19}$$

Symbol of median $\tilde{x}$

* For even terms:-

If data is given as

18, 19, 20, 25, 26, 28

$$\text{Median} = \frac{n\text{th} + \left[\frac{n+1}{2} \right]\text{th term}}{2}$$

total terms $n = 6$ so

$$\text{median} = \frac{3^{\text{th}} + 4^{\text{th}} \text{ term}}{2} = \frac{20 + 25}{2} = 22.5$$

$$\tilde{x} = 22.5$$

Note: "If data is not arranged then first of all arrange it in ascending order."

For grouped data

$$\tilde{x} = l + h \cdot \left(\frac{n - c}{f} \right)$$

median class

l = lower class boundary of $\tilde{x}$ class

h = Size of class interval of $\tilde{x}$ class

f = frequency of median class

n = sum of frequencies = Σf

c = cumulative frequency before the median class

Example #10:-

Date: _____

Sol:-

Time	f	C.B	C.F
118-126	3	117.5-126.5	3
127-135	5	126.5-135.5	8
136-144	9	135.5-144.5	17
145-153	12	144.5-153.5	29
154-162	5	153.5-162.5	34
163-171	4	162.5-171.5	38
172-180	2	171.5-180.5	40

$$\Sigma f = 40$$

$\therefore \Sigma f = 40$ $\therefore \frac{n}{2} = \frac{40}{2} = 20$ in C.F Column

20 lies under 29 so median lies in fourth class, which is the median class -

$$L = 144.5 = \text{L.C boundary}$$

$$h = 9 = \text{size of class interval}$$

$$f = 9 = \text{frequency of median class}$$

$$n = 40 = \text{sum of frequencies}$$

$$C = 17 = \text{C.F before median class}$$

$$\tilde{x} = 144.5 + \frac{9}{2}$$

M T W T F S
b) Arranged in an array:-

119, 124, 126, 128, 132, 135, 135, 135, 138, 138,
140, 140, 142, 142, 144, 144, 145, 145, 146, 146, 147, 147,
148, 149, 150, 150, 152, 153, 154, 156, 157, 158, 161,
163, 164, 165, 175, 178.

$$\tilde{x} = \left(\frac{n}{2}\right)^{\text{th}} + \left(\frac{n}{2} + 1\right)^{\text{th}}$$

$$= \frac{\left(\frac{40}{2}\right)^{\text{th}} + \left(\frac{40}{2} + 1\right)^{\text{th}}}{2} = \frac{20^{\text{th}} + 21^{\text{th}}}{2}$$

$$\tilde{x} = \frac{146 + 146}{2} = 146 \text{ LB}$$

Mode

* For ungrouped data:

locating the maximum frequency

Example 11(a)

First of all arranged data

1, 2, 3, 4, 5, 7, 8, 8, 8, 8, 10, 11, 12, 15, 17,

$$\hat{P} = 8$$

because 8 occurs 4 times

and it is maximum frequency

ii) Mode for discrete frequency distribution:-

The mode exist alongside the highest frequency

Example 12(a)

Size of Shoes x

No of Pair sold (f)

5 1/2

5

15

6 1/2

25

6

50

7 1/2

30

8

20

10

As the highest frequency is 50 so corresponding value 6 1/2 is the modal value of shoes size.

iii) Mode for grouped data:-

The mode

would lie in that class which has the maximum frequency. This class is called the modal class.

$$\hat{X} = l + \frac{f_2 - f_1}{f_1 + f_2} \times h$$

$f_1 + f_2$

l = lower class boundary

f_1 = frequency of class before model class

f_2 = frequency of class ~~after~~ ^{following} model class

h = length of class interval

Example #13:

calculate modal and median heights

Sol:

X	f	C.B	C.F
61	5	59.5-62.5	5
64	18	62.5-65.5	23
67	42	65.5-68.5	65
70	27	68.5-71.5	92
73	08	71.5-74.5	100

$$L = 65.5$$

$$f_1 = 18$$

$$f_2 = 27$$

$$h = 3$$

$$\hat{X} = L + \frac{f_2}{f_1 + f_2} \times h$$

$$= 65.5 + \frac{27}{18+27} \times 3$$

$$\hat{X} = 67.3$$

Empirical Relation b/w mean, median and mode:-

Difference b/w mean and median

$$\frac{\text{Mean} - \text{Median}}{2} = \frac{\text{Median} - \text{mode}}{2}$$

$$2(\text{mean} - \text{median}) = (\text{median} - \text{mode})$$

$$2(\bar{X} - \tilde{X}) = \tilde{X} - \hat{X}$$

$$\hat{X} = \tilde{X} - 2\bar{X} + 2\tilde{\tilde{X}}$$

$$\hat{X} = 3\tilde{X} - 2\bar{X}$$

Geometric Mean:-

$$G_1 = \sqrt[n]{(X_1 \cdot X_2 \cdot X_3 \dots X_n)}$$

If n is too long / large then easiest way:-

$$G_1 = \text{Antilog} \left(\frac{\sum \log X}{n} \right) \rightarrow \text{for ungrouped data}$$

$$\text{and } G_1 = \text{Antilog} \left(\frac{\sum (f \log x)}{\sum f} \right) \rightarrow \text{for group data}$$

Example 15(b)

calculate geometric mean

Sol:-

X	f	log X	f log X
13	2	1.1139	2.2278
14	5	1.1461	5.7305
15	13	1.1761	15.2893
16	7	1.2041	8.4287
17	3	1.2304	3.6912
$\sum f = 30$			$\sum f \log X = 35.3675$

$$G.M = \text{Antilog} \left[\frac{\sum (f \log X)}{\sum f} \right]$$

$$= \text{Antilog} \left[\frac{35.3675}{30} \right] = \text{Antilog} (1.1789) = 15.09$$

Harmonic mean:-

"Reciprocal of the mean of the reciprocals of the numbers"

H = Reciprocal of mean ($\bar{X}$) of reciprocal of values

H = Reciprocal of $\bar{X}$ of $\left[\frac{1}{x_1}, \frac{1}{x_2}, \dots, \frac{1}{x_n} \right]$

$$H = \frac{1}{\frac{1/x_1 + 1/x_2 + 1/x_3 + \dots + 1/x_n}{n}}$$

$$H = \frac{n}{1/x_1 + 1/x_2 + 1/x_3 + \dots + 1/x_n}$$

$$= \frac{n}{\sum 1/x_i} \quad \text{or} \quad \frac{1}{H} = \frac{\sum 1/x_i}{n}$$

If $x_1, x_2, x_3, \dots, x_n$ represents class marks with frequencies $f_1, f_2, \dots, f_n$ then H.M.

$$H = \frac{n}{\sum f_i \left[\frac{1}{x_i} \right]} = \frac{\sum f}{\sum f \left[\frac{1}{x} \right]}$$

Example 16(a):

As the distance travel is same for each day only the speed per kilometer is changes

Days

a

b

1st

400

70 km/hr

2nd

400

90 km/hr

3rd

400

85 km/hr

$$H.M = 3$$

$$\frac{1}{70} + \frac{1}{90} + \frac{1}{85} = 80.73 \text{ kmh}^{-1}$$

Date: _____

Relationship b/w A.M, G.M and H.M

$$A.M > G.M > H.M$$

$$\text{If } (x_1 = x_2 = \dots = x_n)$$

$$\text{then } H.M = G.M = A.M$$

Example # 18:-

consider a sample with data values of 10, 20, 12, 17, 16. Prove $A.M > G.M > H.M$

X	$\log x$	$1/x$
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10	1	0.1
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20	1.3010	0.05
----	--------	------

12	1.079	0.0833
----	-------	--------

17	1.230	0.0588
----	-------	--------

16	1.204	0.0625
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$$\sum X = 75 \quad \sum \log x = 5.814 \quad \sum 1/x = 0.3546$$

$$A.M = \frac{\sum X}{n} = \frac{75}{5} = 15$$

$$G.M = \text{Antilog} \left(\frac{\sum \log x}{n} \right) = \text{Antilog} (1.1628) = 14.547$$

$$H.M = \frac{5}{0.3546} = 14.100$$

$$A.M > G.M > H.M$$

Quartile:

The three points dividing an arranged series of values in four equal parts are known as quartiles and denoted by Q_1, Q_2, Q_3 .

Quartiles for individual observations (ungrouped data)

Q_1 = value of $\left(\frac{n+1}{4}\right)^{\text{th}}$ term of array data

Q_2 = " " $2\left(\frac{n+1}{4}\right)^{\text{th}}$ " " " "

Q_3 = " " $3\left(\frac{n+1}{4}\right)^{\text{th}}$ " " " "

Quartiles for a frequency distribution of discrete data:

Q_1 = value of $\left[\frac{n+1}{4}\right]^{\text{th}}$ item under C.F. Commulative Frequency

Q_2 = " " $2\left(\frac{n+1}{4}\right)^{\text{th}} - \frac{n+1}{2}$ " " "

Q_3 = " " $3\left(\frac{n+1}{4}\right)^{\text{th}}$ " " "

Percentiles:-

The ninety nine points which divide the data set in 100 equal parts and denoted by $P_1, P_2, \dots, P_{99}$.

Percentiles for ungrouped data:-

P_j = value of $j\left(\frac{n+1}{100}\right)$ item of array data

Percentiles for discrete frequency distribution:

P_j = value of $j\left(\frac{n+1}{100}\right)^{\text{th}}$ under C.F.

Example 19(a).-

Date: _____

The weight measurement of 15 onions in grams are given
65, 68, 60, 78, 65, 74, 58, 54, 61, 57, 63, 64, 72, 66, 67

Find three quartiles D_6 , P_{40} and P_{85} .

Sol:-

Arranging $X = 54, 57, 58, 60, 61, 63, 64, 65, 65, 66, 67, 68, 72, 74, 78$

$$Q_1 = \left(\frac{n+1}{4} \right) \text{th term} = \left[\frac{15+1}{4} \right] = 4 \text{th term}$$

$Q_1 = 60$

$$Q_2 = 2 \left(\frac{n+1}{4} \right) \text{th term} = 2 \left(\frac{16}{4} \right) = 8 \text{th term}$$

$Q_2 = 65$

$$Q_3 = 3 \left(\frac{16}{4} \right) = 12 \text{th term} = 68$$

$$D_6 = 6 \left[\frac{n+1}{10} \right] - 6 \left[\frac{16}{10} \right] = 9.6 \text{th}$$

$$= 9 \text{th term} + 0.6 (10 - 9 \text{th term})$$

$$= 65 + 0.6$$

$$D_6 = 65.6$$

$$P_{40} = 240 \left[\frac{n+1}{5 \times 100} \right] = 2 \left[\frac{16}{5} \right] = 6.4 \text{th term}$$

$$= 6 \text{th term} + 0.4 (5 \text{th term} - 4 \text{th term})$$

$$= 63 + 0.4 (61 - 60)$$

$$= 63.4$$

Measure of Dispersion:-

The degree to which numerical data tend to spread on average value is called the dispersion or variation.

Types of measure of Dispersion:

- i) Range (R)
- ii) Quartile Deviation (Q.R)
- iii) Mean Deviation (M.D)
- iv) Standard deviation

Relative measure of Dispersion:

Relative dispersion = $\frac{\text{Absolute Dispersion}}{\text{Average}}$

Range (R)

$$R = X_m - X_o$$

largest observation
smallest observation

Coefficient of range:-

$$\text{Coefficient of range} = \frac{X_m - X_o}{X_m + X_o}$$

Example # 20:-

108, 110, 112, 124, 127, 118, 113

$$\text{Range } R = X_m - X_o = 127 - 108 = 19$$

$$\text{Coefficient of Dispersion} = \frac{X_m - X_o}{X_m + X_o}$$

$$= \frac{19}{235}$$

$$= 0.0808$$

of Quartile deviation and coefficient

Quartile range - $Q.R = (Q_3 - Q_1)$

" deviation - $Q.D = (Q_3 - Q_1)$

coefficient of " " = $\frac{Q_3 - Q_1}{Q_3 + Q_1}$

Example 21(a):

55, 42, 47, 56, 49, 46, 51, 60, 46 are marks of a student. Find median, $Q.D$ and coefficient of $Q.D$

Solution:-

Data array = 42, 46, 46, 47, 49, 51, 55, 56, 60

median $\bar{x} = \left[\frac{n+1}{2} \right]$ th term

= $\left[\frac{10}{2} \right]$ th term = 5th term

$\bar{x} = 49$

$Q_1 = \left[\frac{n+1}{4} \right]$ th item = $\left(\frac{10}{4} \right)$ th term

= 2.5th

= 2th term + $0.5(3^{rd} - 2^{nd})$

= $46 + 0.5(46 - 46)$

= 46

$Q_2 = 2 \left[\frac{n+1}{4} \right] = 2 \left(\frac{10}{4} \right)$ = 5th term

= 49

$$Q_3 = 3\left[\frac{n+1}{4}\right] \text{th item} = 3\left[\frac{10}{4}\right] \text{th item}$$

$$= 7.5 \text{th item}$$

$$= 7\text{th} + 0.5(8^{\text{th}} - 7^{\text{th}})$$

$$= 55 + 0.5(56 - 55)$$

$$Q_3 = 55.5 \approx 56$$

$$O.D = Q_3 - Q_1 = 56 - 46 = 4.5$$

$$\text{Coefficient of O.D} = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{4.5}{101}$$

$$= 0.0445$$

Mean deviation and coefficient of M.D.

$$\rightarrow M.D = \bar{X} = \frac{\sum |X - \bar{X}|}{n}$$

mean deviation about mean n

other names: average deviation, Absolute mean deviation

$$\rightarrow M.D \hat{X} = \frac{\sum |X - \hat{X}|}{n}$$

Mean deviation about mode n

$$\rightarrow M.D \tilde{X} = \frac{\sum |X - \tilde{X}|}{n}$$

mean deviation about median n about situation $(\bar{X}, \hat{X}, \tilde{X})$

for grouped data = $\frac{\sum f |X - \text{Average}|}{N}$

$$\text{Coefficient of M.D (about mean)} = \frac{M.D \bar{X}}{\bar{X}}$$

$$\text{Coefficient of M.D (about mode)} = \frac{M.D \tilde{X}}{\tilde{X}}$$

" " (about median) = M.D.
 $\bar{x}$

Example 22(a)

find the mean deviation

12, 6, 7, 3, 15, 10, 18, 5

$$\bar{x} = \frac{12+6+7+3+15+10+18+5}{8} = \frac{76}{8} = 9.5$$

now M.D about mean

$$M.D.\bar{x} = \frac{\sum |x - \bar{x}|}{n}$$

x	$x - \bar{x}$	$ x - \bar{x} $
-----	---------------	-----------------

12	2.5	2.5
----	-----	-----

6	-3.5	3.5
---	------	-----

7	-2.5	2.5
---	------	-----

3	-6.5	6.5
---	------	-----

15	5.5	5.5
----	-----	-----

10	0.5	0.5
----	-----	-----

18	8.5	8.5
----	-----	-----

	-4.5	4.5
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	0	$\Sigma = 34$
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$$M.D.\bar{x} = \frac{34}{8} = 4.25$$

Example 22(b)

Marks	f	x	fx	$ x - \bar{x} $	$f x - \bar{x} $
0-9	2	4.5	9	41.67	83.34
10-19	3	14.5	43.5	31.67	95.01
20-29	8	24.5	196.0	21.67	173.36
30-39	24	34.5	828.0	11.67	280.08
40-49	21	44.5	934.5	1.67	35.07
50-59	40	54.5	2180	8.33	333.2
60-69	11	64.5	709.5	18.33	201.63
70-79	5	74.5	372.5	28.33	141.65
$\Sigma f = 120$		$\Sigma fx = 5540$		$\Sigma f x - \bar{x} = 1353.36$	

$$\bar{x} = \frac{\Sigma fx}{\Sigma f} = \frac{5540}{120} = 46.17$$

$$M.D = \frac{\Sigma f|x - \bar{x}|}{n} = \frac{1353.36}{120} = 11.278$$

Standard Deviation:-

The standard deviation is a measure of the amount of variation of data set

$$S = \sqrt{\frac{\Sigma (x - \bar{x})^2}{n}} \rightarrow \text{for sample standard deviation}$$

$$\sigma = \sqrt{\frac{\Sigma (x - \mu)^2}{N}} \rightarrow \text{for population " "}$$

$$s = \frac{\sum f(x - \bar{x})^2}{\sum f}$$

Grouped frequency

$$\sigma = \sqrt{\frac{\sum f(x - \bar{x})^2}{\sum f}}$$

Example 23(a)

12, 6, 7, 3, 15, 10, 18, 5

$$\bar{x} = \frac{12 + 6 + 7 + 3 + 15 + 10 + 18 + 5}{8} = 9.5$$

x	$x - \bar{x}$	$(x - \bar{x})^2$
12	2.5	6.25
6	-3.5	12.25
7	-2.5	6.25
3	-6.5	42.25
15	5.5	30.25
10	0.5	0.25
18	8.5	72.25
5	-4.5	20.25

$$\sum (x - \bar{x})^2 = 190$$

$$s = \frac{\sum (x - \bar{x})^2}{n} = \frac{190}{8} = 23.75$$

$$= 4.87$$

Example 23(b)

MTWTFSS

Date: _____

Height	f	^{class mark} x	fx	$x - \bar{x}$	$(x - \bar{x})^2$	$f(x - \bar{x})^2$
60-62	5	61	305	-6.45	41.60	208.00
63-65	18	64	1152	-3.45	11.90	214.20
66-68	42	67	2814	-0.45	0.20	8.50
69-71	27	70	1890	2.55	6.50	175.50
72-74	8	73	584	5.55	30.80	246.40

$$\sum f = 100$$

$$\sum fx = 6745$$

$$\sum f(x - \bar{x})^2 = 852.75$$

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{6745}{100} = 67.45 \text{ inches}$$

$$s = \sqrt{\frac{\sum f(x - \bar{x})^2}{\sum f}} = \sqrt{\frac{852.75}{100}}$$

$$= 2.92 \text{ inches}$$

